In this image are the BlinkM MinM smart LED and the DS1307 RTC breakout board from SparkFun Electronics. Unfortunately, neither of these devices comes ready to plug right into the breadboard. In the case of the BlinkM MinM, it includes the four-pin male pin header shown in the image, while the SparkFun device requires a separate purchase of the pin headers needed. Pin headers are kind of like the glue of electronics prototyping, so having a few extra on hand is always a good idea. We will make the assumption here that to complete some of the projects discussed earlier, you might want to know how to go about soldering these pins to these breakout boards.

To get started soldering you'll need to pick up a few things. The first thing, of course, is a soldering iron. On the low end, a $10–$20 pen-style soldering iron in the 30-watt ballpark is just fine for learning how to solder. A soldering stand and a sponge to clean the soldering iron tip are also good ideas. We like to use our Weller WES-51 adjustable soldering station, but other higher-end stations that run about $100 are also a good choice if you find that you'll be doing quite a bit of soldering. You will also need some solder. As long as you avoid the solder for copper pipes that you might find at the hardware store, you will probably be fine. We find .03" diameter 60/40 tin/lead rosin core solder works great for general, personal-use soldering. Other sizes and types of solders, including those that are lead-free, are also available and will work fine. It can also be tricky to hold four different things—the iron, solder, and two parts—while soldering, so some form of clamp or helper is also a good idea to have on hand.

A quick note here about safety. We are indeed recommending lead-based solder for personal use because of its ease of use and it is generally nicer to our irons. The fumes that come from soldering are mostly the rosin inside the solder burning off, but you do not necessarily want to make a habit of breathing it. Lead-free solder is not only harder on the iron, but it also uses far more toxic rosin, so you will want to seriously ventilate when using this type of solder, if you choose to use it. Either way, you should always wash your hands after handling any kind of solder.

Okay, before we start soldering, we need to get the iron good and hot. We also want a nice, clean, and shiny tip at the end of our iron. We can get this by applying a little bit of solder to the tip of our iron and then wipe the tip across a damp sponge. This is a process called tinning. It is also a good idea to clean the tip before and after each use. A clean tip will make the heat transfer better and the solder flow more smoothly. With a properly tinned tip, it's now time to solder. Figure 12-17 provides the illustrated and abbreviated version of this process.